

Lecture 1: Real Numbers

AMA3707 Real Analysis

1 Numbers

Definition 1.1 (Natural Numbers). [def:natural-number] The *natural numbers* are the positive integers:

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

Convention 1.1 (Natural Numbers Exclude Zero). [conv:natural-numbers-exclude-zero] $0 \notin \mathbb{N}$.

Definition 1.2 (Integers). [def:integer] The *integers* are the whole numbers, including the positive integers, the negative integers, and zero:

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

Definition 1.3 (Rational Numbers). [def:rational-numbers] A real number is called *rational* if it can be expressed as a quotient of two integers. The set of rational numbers is

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Proposition 1.1 (Unique Reduced Form). [prop:unique-reduced-rational-form] For every $x \in \mathbb{Q}$, there exists a unique pair $p, q \in \mathbb{Z}$ such that

$$x = \frac{p}{q}, \quad q > 0, \quad \gcd(|p|, q) = 1.$$

Remark 1.1. [rem:number-set-inclusions] Under the usual identifications, $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q}$.

2 Fields

Definition 2.1 (Closed Operation). [def:closed-operation] An operation $*$ is said to be *closed* on a set A if for all $a, b \in A$, we have $a * b \in A$.

Definition 2.2 (Field). [def:field] A *field* is a set F equipped with addition and multiplication such that both operations are multiplication are

- closed operations on F ,
- associative, meaning for all $a, b, c \in F$, we have $(a+b)+c = a+(b+c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$,
- commutative, meaning for all $a, b \in F$, we have $a + b = b + a$ and $a \cdot b = b \cdot a$,
- have identity elements 0 and 1,
- each element has an additive inverse, such that for each $a \in F$, there exists $-a \in F$ with $a + (-a) = 0$,
- each non-zero element has a multiplicative inverse, such that for each $a \in F$ with $a \neq 0$, there exists $a^{-1} \in F$ with $a \cdot a^{-1} = 1$

Proposition 2.1 ($\mathbb{Q}$ Is a Field). [prop:rational-numbers-are-a-field] The rational numbers with their usual addition and multiplication form a field.

Remark 2.1 ($\mathbb{N}$ and $\mathbb{Z}$ Are Not Fields). [rem:natural-and-integers-not-fields] The natural numbers and the integers are not fields because they do not have multiplicative inverses for every non-zero element.

For a counter example, consider the integer 2. There is no integer x such that $2 \cdot x = 1$

3 Ordered Set

Definition 3.1 (Ordered Set). [def:ordered-set] A set F is *ordered* if it is transitive and reflexive and

$$\forall x, y \in F, \text{ either one of the following happens } x \leq y, x = y, \text{ or } y \leq x$$

It is also called partial ordered.

Definition 3.2 (Transitive). [def:transitive] An ordering on a set is *transitive* if

$$\forall x, y, z \in F, \text{ if } x \leq y \text{ and } y \leq z, \text{ then } x \leq z$$

Definition 3.3 (Reflexive). [def:reflexive] An ordering on a set is *reflexive* if

$$\forall x \in F, x = x$$

4 Irrational Numbers

Lemma 4.1 (Parity of a Square). [lem:square-parity] For every $n \in \mathbb{Z}$,

$$n^2 \text{ is even} \iff n \text{ is even},$$

and equivalently,

$$n^2 \text{ is odd} \iff n \text{ is odd}.$$

Proposition 4.1 (No Rational Square Root of 2). [prop:no-rational-square-root-of-two] There is no $q \in \mathbb{Q}$ such that $q^2 = 2$.

Proof. Suppose, for contradiction, that some $q \in \mathbb{Q}$ satisfies $q^2 = 2$. According to the proposition 1.1, and $q \in \mathbb{Q}$

$$\exists p, r \in \mathbb{Z}, q = \frac{p}{r}, r > 0, \gcd(|p|, r) = 1$$

Then we have

$$\left(\frac{p}{r}\right)^2 = 2 \implies p^2 = 2r^2$$

Since p^2 is even, the square-parity lemma implies that p is even. Therefore,

$$\exists k \in \mathbb{Z}, p = 2k.$$

So,

$$(2k)^2 = 2r^2 \implies 2k^2 = r^2$$

By the same argument, r is also even. Meaning that both p and r are even, which contradicts the assumption that $\gcd(|p|, r) = 1$.

So, it contradicts the assumption that $q \in \mathbb{Q}$ satisfies $q^2 = 2$. Therefore, there is no $q \in \mathbb{Q}$ such that $q^2 = 2$. $\square$

Remark 4.1 (Larger Number System). [rem:sqrt-two-larger-number-system] Within $\mathbb{Q}$, the proposition only says that the equation $q^2 = 2$ has no solution $q \in \mathbb{Q}$. Writing $\sqrt{2} \notin \mathbb{Q}$ presupposes a larger number system in which $\sqrt{2}$ exists.

Remark 4.2 (Completeness). [rem:rational-incompleteness] The existence, in a larger number system, of some $x \notin \mathbb{Q}$ satisfying $x^2 = 2$ does not by itself prove that $\mathbb{Q}$ is incomplete. Incompleteness must be demonstrated internally [TODO Pf link to completeness axiom].